π^0-based calibration of the ePIC backward electromagnetic calorimeter

Validation with 18×275 GeV DIS and low-Q^2 photoproduction samples

EEEMCal simulation study

27 September 2026

Abstract

We study a $\pi^0 \to \gamma\gamma$ calibration procedure for the ePIC backward electromagnetic calorimeter using two independent 18×275 GeV simulation samples. The first is neutral-current DIS with $Q^2 > 1$ GeV^2; the second is the low-Q^2 sample stored in the SIDIS production, corresponding here to photoproduction kinematics with $0 < Q^2 < 1$ GeV^2. The same photon selection, training/validation split and gain-solving procedure are used in both samples. This makes the comparison useful as a validation of the method over very different event samples and statistical regimes.

The DIS sample contains about 5.0 million reconstructed events and 45,385 training pairs in the π^0 calibration window. It is sufficient to determine gains over essentially the full calorimeter, but some outer crystals remain weakly constrained. The min20 solution fits 2,736 crystals and moves the held-out peak from 128.4 to 134.6 MeV. The relative mass width changes from 3.87% to 3.93%, corresponding to a small 1.4% degradation.

The photoproduction sample contains about 48 million reconstructed rows and 703,165 training pairs in the calibration window, about 15.5 times more than DIS. All 2,740 crystals receive a contribution above the 5% shower-energy threshold in at least 217 calibration events, and the min20 and min10 solutions become identical. The gain map is much more stable under independent half-sample fits, and the held-out relative mass width improves from 3.43% to 3.31%, a 3.5% relative improvement. The photons entering the photoproduction calibration are slightly softer than those in DIS: the mean reconstructed photon energy is 1.11 GeV instead of 1.21 GeV, with most of the difference in the high-energy tail. These two studies validate the numerical π^0 calibration procedure in realistic ePIC samples; a full reconstruction feedback test remains the next step before the gains can be regarded as a detector-level calibration.

1 Motivation and strategy

The purpose of this study is to establish whether a π^0-based channel calibration can be used reliably for the backward electromagnetic calorimeter. The basic idea is standard: the reconstructed two-photon mass provides a common reference, and the crystal gains are adjusted so that the π^0 peak is brought to its nominal value. What needs to be checked for the ePIC application is whether the available event samples constrain all channels well enough, whether the numerical solution is stable, and whether the correction improves an independent sample rather than only the events used to determine the gains.

We therefore apply exactly the same procedure to two samples with very different characteristics. The DIS sample has $Q^2 > 1$ GeV^2 and relatively modest π^0 statistics. The second sample comes from the low-Q^2 SIDIS production and covers the photoproduction region, $0 < Q^2 < 1$ GeV^2; it provides roughly an order of magnitude more reconstructed events and more than an order of magnitude more pairs in the calibration window. The DIS study tests the method close to the minimum useful statistical regime, while the photoproduction sample shows how the same procedure behaves when every crystal is well illuminated.

All results below use the final relaxed selection with $E_\gamma \geq 0.2$ GeV. Earlier pilot selections and exploratory studies of the minimum per-crystal statistics are not mixed into the main comparison.

2 Common event selection and calibration method

2.1 Candidate selection

For each reconstructed EEEMCal cluster, the compact analysis sample stores the cluster energy and position, the selected event vertex, the crystal cell IDs and the energy assigned to each crystal. The assigned crystal energies are used directly. For shower j in event i,

$$s_{ji} = \sum_k c_{jik}, \tag{1}$$

where c_{jik} is the contribution from crystal k.

The same pair selection is used for DIS and photoproduction. Photon candidates must be neutral and have $E_\gamma \geq 0.2$ GeV. We require an energy asymmetry $|E_1 - E_2|/(E_1 + E_2) < 0.8$, an assigned-energy overlap below 0.2, and different seed crystals. If more than one pair is available in an event, the pair with the largest $E_1 + E_2$ is retained. Neither truth information nor the two-photon invariant mass enters this choice.

Before fitting the π^0 peak, events are divided deterministically into an approximately 80/20 training/held-out split using the file and event identity. The training spectrum is fitted with a Gaussian signal plus a smooth exponential background. A $\pm 2\sigma$ interval around the fitted peak is then fixed and used for the gain solve. The held-out sample is kept completely independent and is evaluated with the broad selection $E_\gamma \geq 0.2$ sample, without imposing the training mass window.

2.2 Gain determination

The gain calculation follows the bootstrap method of Ref. [1]. For an event with two reconstructed photons,

$$m_i^2 = 2p_{1i}p_{2i}(1 - \cos\gamma_i), \tag{2}$$

where γ_i is the reconstructed opening angle. In the fixed-shower approximation, applying a relative gain G_k to crystal k changes shower j according to

$$p_{ji}(G) = p_{ji}^{(0)} \frac{\sum_k G_k c_{jik}}{\sum_k c_{jik}}. \tag{3}$$

For a small multiplicative change $G_k \to G_k(1 + \epsilon_k)$,

$$J_{ik} = \frac{\partial m_i^2}{\partial \epsilon_k} = m_i^2 \left(\frac{c_{1ik}}{s_{1i}} + \frac{c_{2ik}}{s_{2i}} \right). \tag{4}$$

The two shower normalizations are kept separate, and a crystal shared by the two showers contributes through both terms. With $r_i = m_i^2 - m_{\pi^0}^2$, the linearized update is built from $C = J^T J$, $D = -J^T r$, $L = J^T \mathbf{1}$ and $B = \sum_i r_i$, with the same global constraint in both analyses. Modes below 10^{-3} of the largest eigenvalue are discarded. Gains are updated multiplicatively, with a maximum relative step of 0.1 and the same positivity and convergence safeguards in the two samples.

To quantify the statistics available for each crystal, we count the selected calibration events in which that crystal carries at least 5% of either photon-shower energy. The main results require at least 20 such events per crystal (min20). A min10 solution is also kept as a cross-check.

3 Simulation samples

The two datasets are both from the 25.10.0 `epic_craterlake` reconstruction campaign at 18×275 GeV, but they cover complementary Q^2 regions.

The DIS sample is the neutral-current production with $Q^2 > 1$ GeV^2. The reconstructed files are taken from

```
/volatile/eic/EPIC/RECO/25.10.0/epic_craterlake/DIS/NC/18x275/minQ2=1/
```

The complete sample contains 11,379 reconstructed files and 4,967,376 events.

The second sample is the low-Q^2 production stored under the SIDIS directory. In this report it is referred to as the SIDIS/photoproduction sample and corresponds to $0 < Q^2 < 1$ GeV^2. The reconstructed files are taken from

```
/volatile/eic/EPIC/RECO/25.10.0/epic_craterlake/SIDIS/pythia6-eic/1.0.0/18x275/q2
_0to1/
```

Seventeen extraction tasks encountered truncated input files. Rather than mix partial outputs with complete ones, all products associated with those tasks were removed. The final sample therefore contains 71,807 of the 73,507 files in the original manifest, or 97.7%, corresponding to 48,314,330 reconstructed rows.

Table 1: Size of the two samples after the common Selection-B analysis.

Quantity	DIS ($Q^2 > 1$)	SIDIS/photoproduction ($0 < Q^2 < 1$)
Reconstructed files	11,379	71,807
Reconstructed events/rows	4,967,376	48,314,330
Selection-B pairs, total	293,182	5,398,486
Training pairs	234,523	4,319,942
Held-out pairs	58,659	1,078,544
Training-window pairs	45,385	703,165

For DIS, using $\sigma_{\text{gen}} \simeq 7.4 \times 10^5$ pb gives an equivalent luminosity of about 6.7 pb^{-1}, or 1.9 h at 10^{33} cm^{-2}s^{-1} and 11 min at 10^{34} cm^{-2}s^{-1}. For the SIDIS/photoproduction sample, the run-by-run sum $\sum_r N_{\text{RECO rows},r}/\sigma_r$ is 0.90 nb^{-1}. This corresponds to 0.90 s at 10^{33} cm^{-2}s^{-1} or 0.090 s at 10^{34} cm^{-2}s^{-1}.

4 Training peak and calibration sample

The uncalibrated π^0 peak is already clearly visible in both training samples. Table 2 summarizes the fits used to define the calibration windows.

Table 2: Uncalibrated training peak and mass window used in the gain solve. Fit uncertainties are the local statistical uncertainties returned by the fit.

Quantity	DIS	SIDIS/photoproduction
Training peak [MeV]	128.51 ± 0.03	128.41 ± 0.01
Training σ [MeV]	4.95 ± 0.03	4.40 ± 0.01
Calibration window [MeV]	118.6–138.4	119.6–137.2
Pairs inside window	45,385	703,165
Fitted window purity	78.5%	77.8%

The fitted purities are almost the same, so the main difference is statistical. The photoproduction calibration uses 15.5 times more pairs. Its raw peak is also narrower: the fitted σ is 4.40 MeV compared with 4.95 MeV in DIS. This difference is already present before applying any channel gains and therefore reflects differences between the selected event samples rather than the calibration procedure itself.

5 Crystal coverage and gain solution

The larger photoproduction sample changes the per-crystal statistics quite dramatically. In DIS, all 2,740 crystals appear above the 5% shower-energy threshold in at least 10 calibration events, 2,736 do so in at least 20 events, 2,505 in at least 50, and 1,167 in at least 100. In the SIDIS/photoproduction sample every crystal passes that threshold in at least 217 calibration events; the median count is about 1,100 and the maximum about 7,900.

The corresponding min20 solutions are listed in Table 3. Both matrices are full rank over their active channels, but the high-statistics sample is better conditioned and the fitted gains occupy a much narrower range.

Table 3: Main properties of the min20 gain solutions.

Quantity	DIS min20	SIDIS/photoproduction min20
Fitted / unfitted crystals	2736 / 4	2740 / 0
Retained modes, initial → final	2736 → 2736	2740 → 2740
Unresolved modes	0	0
Condition number, initial → final	261 → 251	161 → 157
Convergence	5 updates	4 updates
Gain minimum	0.913	1.014
Gain Q25	1.035	1.047
Gain median	1.051	1.054
Gain Q75	1.068	1.061
Gain maximum	1.244	1.123

For photoproduction, min20 and min10 contain exactly the same 2,740 channels and give exactly the same saved gain table. This is useful in itself: once the illumination is sufficiently high, the choice between these two minimum-event thresholds becomes irrelevant. In DIS the four additional channels admitted by min10 have little effect on the global mass response, but they are much less well constrained individually.

It is also useful to look at the coefficients channel by channel rather than only through the summary numbers in Table 3. For the figures below we use the full-detector min10 solution in DIS, because it contains all 2,740 crystals. In the SIDIS/photoproduction sample the min10 and min20 maps are exactly identical, so the plotted coefficients are also the primary min20 solution used elsewhere in the report. This choice allows a direct visual comparison of the two complete detector maps without changing the main min20 comparison.

The difference in spread is even clearer in Fig. 2, where the two coefficient sets are overlaid using the same crystal ordering. In DIS, the full-rank min10 solution contains a number of isolated excursions, including 62 channels whose coefficient changes by at least 10% in one of the training-half fits. In photoproduction the coefficients form a much tighter band; only three channels change by more than 5%, and none changes by more than 10%.

The two-dimensional maps show where these variations occur in the detector. No smoothing or spatial interpolation is applied: each square corresponds to the coefficient fitted for one crystal. Figure 3 uses the same coefficient scale for the two samples, so equal colors correspond to equal gain corrections. On this common scale, the larger DIS excursions (mostly near the outer part of the calorimeter, where the calibration statistics are lowest) stand out directly, while the photoproduction coefficients occupy a much narrower interval.

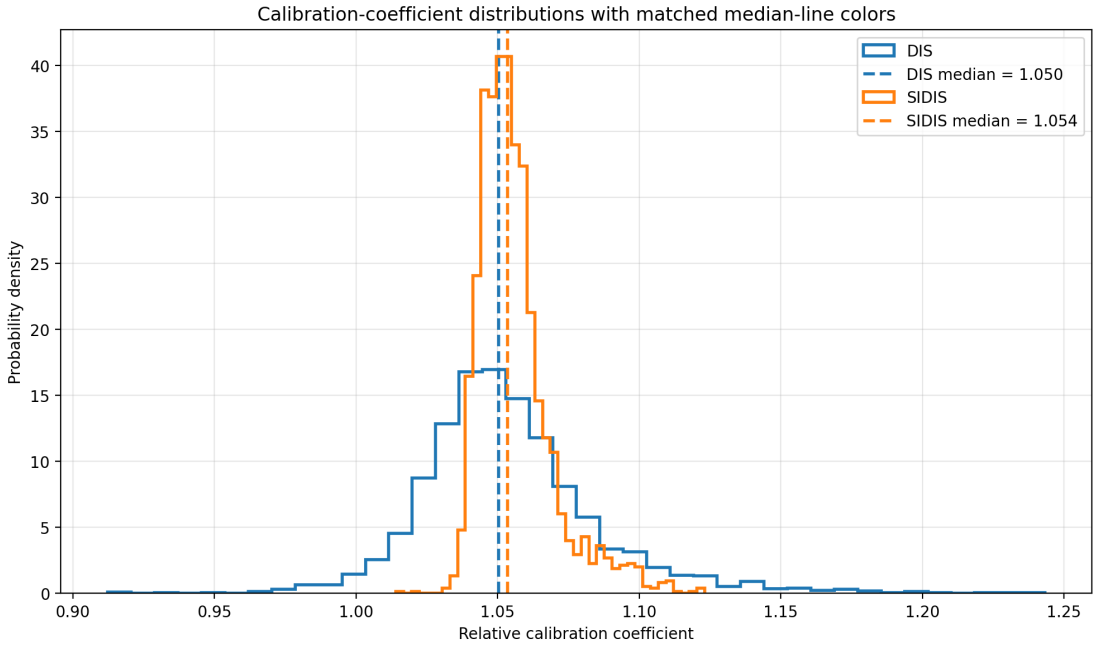


Figure 1: Normalized distributions of the fitted relative calibration coefficients, overlaid for the DIS min10 solution and the SIDIS/photoproduction solution (for which min10=min20). The dashed vertical lines show the corresponding medians, using the same color convention as the histograms. Both samples require an overall correction of about 5%, but the DIS coefficients have much broader tails, whereas the high-statistics photoproduction coefficients are concentrated in a much narrower interval.

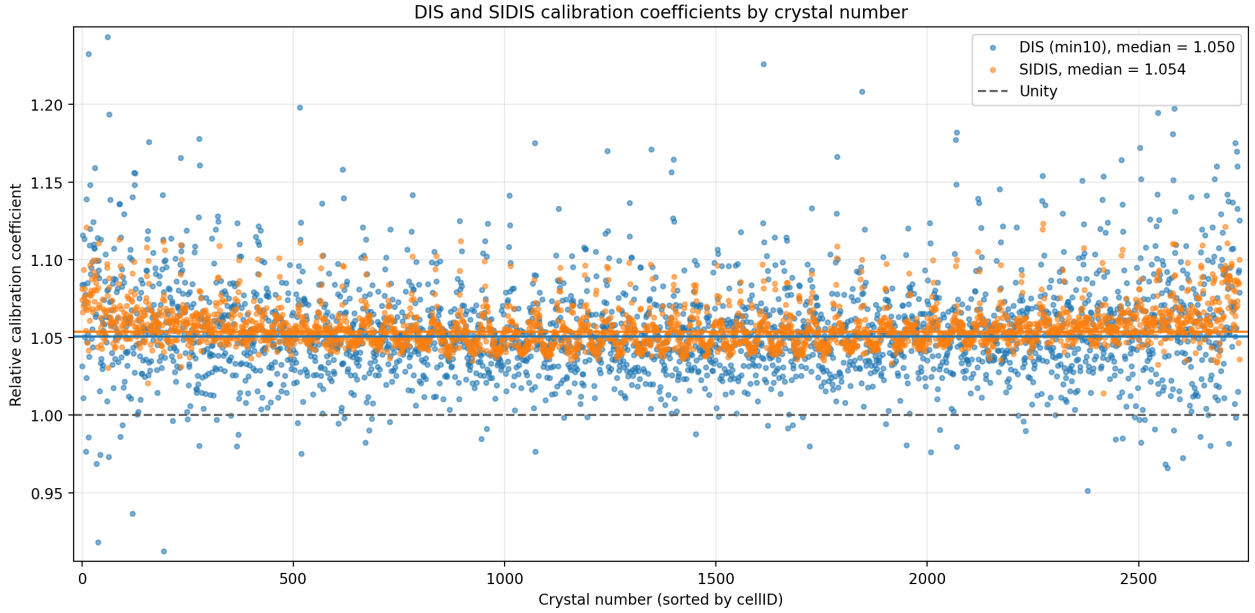


Figure 2: Relative calibration coefficient versus crystal number, with crystals ordered by increasing cellID, shown in a single overlaid view. The DIS min10 coefficients are shown in blue and the SIDIS/photoproduction coefficients in orange. Horizontal lines mark unity and the median value for each sample. The broader vertical spread and more frequent excursions of the DIS coefficients are immediately visible, while the photoproduction coefficients form a much tighter band around their median.

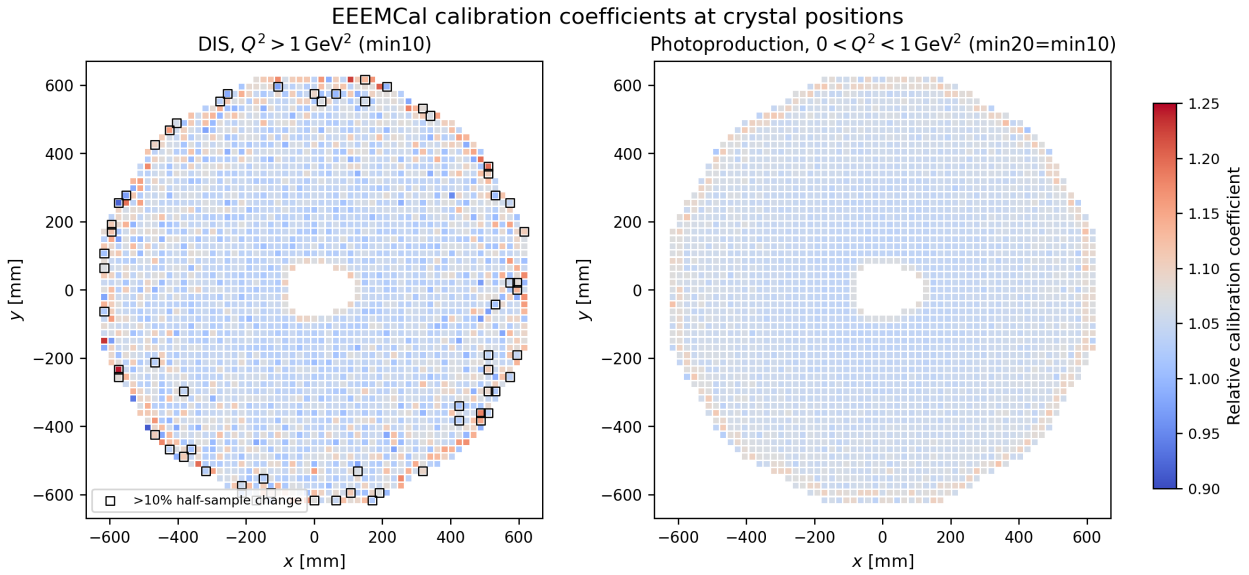


Figure 3: Two-dimensional maps of the fitted relative calibration coefficients at the actual crystal positions. The DIS min10 map (left) and the SIDIS/photoproduction min20=min10 map (right) use the same linear color scale, from 0.90 to 1.25, and a shared color bar. Black outlines in the DIS panel mark channels whose coefficient changes by at least 10% in one of the two training-half fits. The common scale makes the broader coefficient spread in DIS directly comparable with the more uniform photoproduction solution.

6 Stability of the calibration coefficients

A useful test of the gain determination is to repeat it on two independent halves of the training sample. For each crystal we compare the two half-sample results with the full-sample gain and quote the larger relative change. This is not a physical gain uncertainty, but it gives a direct measure of how strongly the data constrain an individual coefficient.

Table 4: Reproducibility of the min20 gains under independent training-half fits.

Quantity	DIS	SIDIS/photoproduction
Median worst-half change	2.1%	0.40%
90th percentile	5.8%	1.2%
Channels below 5%	2351 / 2736	2737 / 2740
Channels below 10%	2679 / 2736	2740 / 2740

The difference is large. In the photoproduction sample, all but three channels agree with the full fit within 5%, and every channel is within 10%. In DIS the least stable coefficients are concentrated near the outer edge of the calorimeter, where the π^0 illumination is lowest. This behavior is exactly what one would expect if limited per-crystal statistics are an important limitation of the lower-statistics fit.

As an independent numerical check, both analyses pass the fixed-shower 5% injected-gain closure test. The relative-error RMS / maximum are 3.6×10^{-8} / 5.6×10^{-7} for DIS and 4.0×10^{-9} / 2.1×10^{-8} for photoproduction. The closure confirms that the solver recovers a known gain pattern within the fixed-shower model; it should not be confused with a statement about detector-model or reconstruction uncertainties.

7 Validation on the held-out sample

The most important test is the response of events that were never used to determine the mass window or the gains. Table 5 shows the result for the min20 solutions.

Table 5: Held-out π^0 response before and after applying the min20 gains.

	DIS		SIDIS/photoproduction	
Quantity	Before	After	Before	After
Peak [MeV]	128.4	134.6	128.4	134.8
σ [MeV]	4.97	5.29	4.40	4.46
σ/μ	3.87%	3.93%	3.43%	3.31%

Both gain maps correct the mass scale by about 5%. In DIS, the fractional width increases slightly, by 1.4%. The photoproduction sample behaves differently: the absolute width increases from 4.40 to 4.46 MeV as the energy scale moves upward, but the more relevant fractional width improves by 3.5%. If the original photoproduction spectrum were simply rescaled to the corrected peak without any improvement in resolution, one would expect $\sigma \simeq 4.62$ MeV; the observed 4.46 MeV is appreciably smaller.

This comparison is an important validation of the method. The small width degradation seen in DIS is not an intrinsic consequence of the gain formalism: the same algorithm, with the same settings, improves the held-out fractional resolution when the calibration coefficients are more strongly constrained. At the same time, the two physics samples are not identical, so the comparison does not prove that statistics alone explains the difference.

8 Photon-energy comparison

The two calibration samples are rather similar at low energy. The main difference is a somewhat longer high-energy tail in DIS. On average, the photoproduction photons are 8.3% lower in energy; the median differs by 4.2%, while the 90th percentile is about 12% lower. The energy asymmetry is essentially the same in the two samples.

Table 6: Photon-energy properties of the training pairs inside the calibration mass windows.

Quantity	DIS	SIDIS/photoproduction
Mean photon energy [GeV]	1.21	1.11
Median photon energy [GeV]	0.88	0.84
Photon-energy standard deviation [GeV]	1.04	0.90
90th percentile [GeV]	2.54	2.22
Mean E_{low} [GeV]	0.78	0.72
Mean E_{high} [GeV]	1.64	1.50
Mean $E_1 + E_2$ [GeV]	2.42	2.22
Median asymmetry	0.305	0.303

Before applying the π^0 mass window, the two productions are much more different: the mean photon energy is 1.70 GeV in DIS and 0.93 GeV in the SIDIS/photoproduction sample. The π^0 selection therefore picks more comparable photon populations out of the two underlying event samples. In particular, the narrower raw peak and the improved calibrated width in photoproduction cannot be attributed to having higher-energy photons; the photons entering that calibration are actually slightly softer.

9 DIS and photoproduction comparison

The two analyses give a consistent picture of how the procedure behaves. The DIS sample is already large enough to establish a detector-wide solution, and it demonstrates that the method can be run with only a few million reconstructed events. Its weakness is not a loss of matrix rank but the limited statistics in the outer crystals. This is visible directly in Figs. 1–3: the DIS coefficient distribution has

longer tails, the overlaid crystal-by-crystal comparison shows several larger excursions, and the largest fluctuations occur mainly near the calorimeter boundary. The half-sample test confirms that these are also the least reproducible coefficients.

The low-Q^2 sample moves the calibration into a much more favorable statistical regime. Every channel is well populated, the condition number is smaller, min10 and min20 become the same problem, and the half-sample gains are stable at the percent level or better. Most importantly, the gain map improves the fractional π^0 width in the independent validation sample. Together with the successful injected-gain closure, this gives strong evidence that the numerical calibration procedure is working as intended.

10 Remaining validation

The present gain fit keeps the shower composition and cluster geometry fixed while the crystal energies are varied. This approximation is what makes the large calibration solve practical, and the numerical tests above show that it is internally consistent. The final detector-level validation, however, must apply the gain table in EICrecon and rerun the reconstruction on an independent event sample. That will test whether changes in RecHit energies alter cluster membership, energy sharing, cluster positions or pair selection in a way that changes the result.

The half-sample variations quoted here should also be viewed as reproducibility tests rather than as physical gain uncertainties, and the fitted changes in mass width do not yet include a full background-

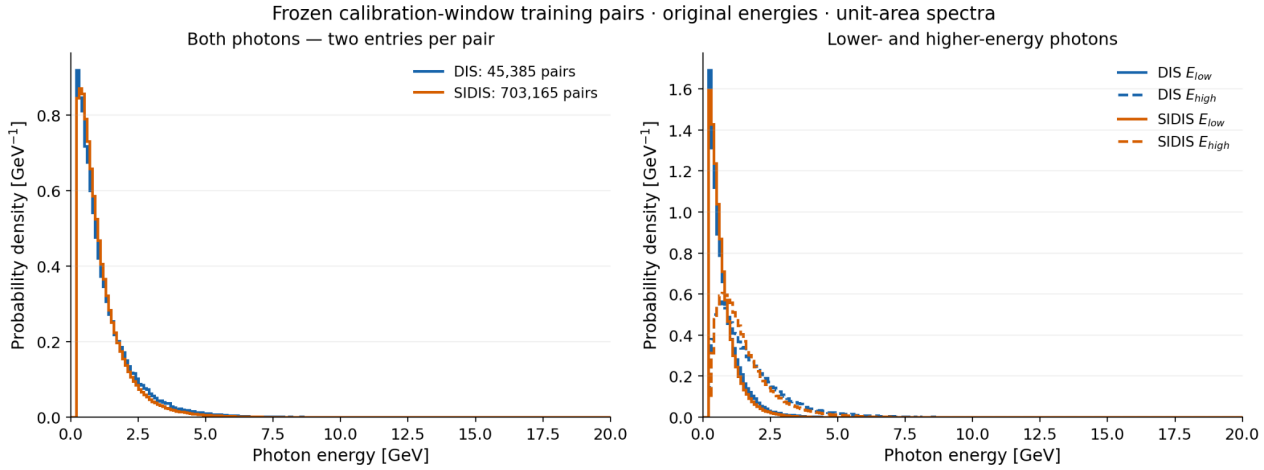


Figure 4: Normalized photon-energy spectra for the training pairs used in the two gain fits. Left: both photons combined. Right: lower- and higher-energy photons shown separately. The DIS sample has a modestly longer high-energy tail.

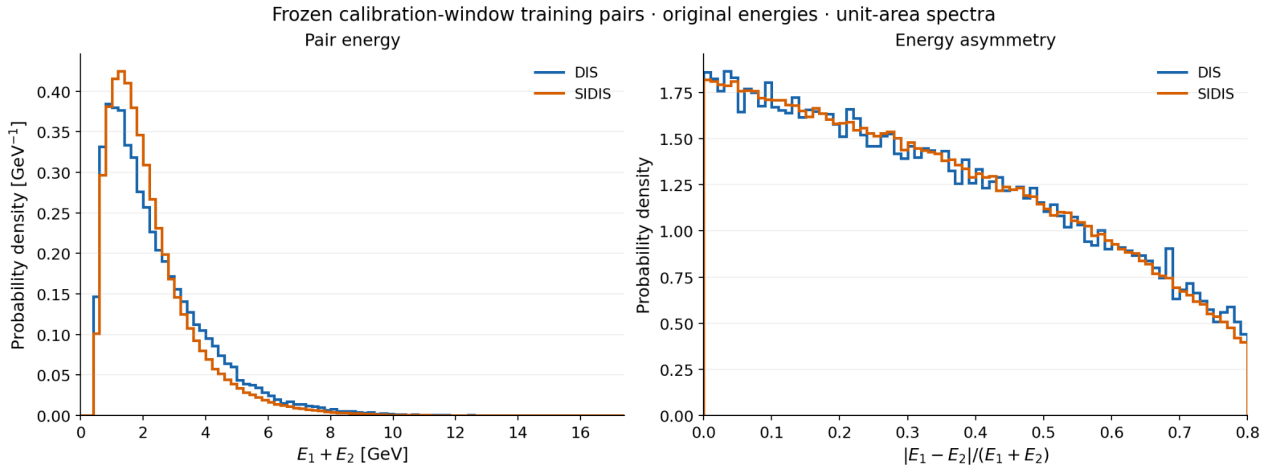


Figure 5: Pair-energy and energy-asymmetry distributions for the same calibration samples. The pair-energy spectrum is slightly softer in photoproduction, whereas the asymmetry distributions are almost indistinguishable.

model or paired statistical systematic study. The calibration-window samples contain background and are not truth-selected π^0 decays. For these reasons the present gains remain candidate calibration constants rather than final detector constants.

11 Conclusion

The DIS and low-Q^2 studies together provide a useful validation of the π^0 calibration procedure for EEEMCal. With $Q^2 > 1$ GeV^2, the DIS sample already gives essentially complete calorimeter coverage and corrects the π^0 mass scale, although the least illuminated channels show appreciable gain variation and the held-out fractional width increases slightly. In the $0 < Q^2 < 1$ GeV^2 SIDIS/photoproduction sample, the much larger π^0 statistics constrain all 2,740 channels strongly. The gain solution is better conditioned, highly reproducible under sample splitting, and improves the held-out fractional mass width by 3.5%.

References

[1] R. T. Jones et al., "A bootstrap method for gain calibration and resolution determination of a lead-glass calorimeter," *Nucl. Instrum. Meth. A* **566** (2006) 366–374, doi:10.1016/j.nima.2006.07.061.